

National University of Singapore
School of Computing

Semester 1, AY2026-27

CS4246/CS5446

Reinforcement Learning and Sequential Decision Making

Issued: 26 Aug 2026

Tutorial 2 Week 4: Rational Decision Making

Guidelines

You may discuss the content of the questions with your classmates. But everyone should work on and be ready to present ALL the solutions.

Problem 1: Utility Matchmaker: The Case of the Mysterious Preferences

Part A. Consider the following lotteries, where $[p, x]$ denotes outcome x with probability p :

$$L_1 = [1, 1], \quad L_2 = [0.5, 2; 0.5, 0], \quad L_3 = [1, 2].$$

Four students have expressed their preferences over these lotteries:

- Adam is indifferent between L_2 and L_1 .
- Bing prefers L_1 to L_2 .
- Candy is indifferent between L_3 and L_2 .
- Diana prefers L_2 to L_1 .

Match each student with a utility function that is consistent with their stated preferences (each student has a different one):

(a) $U(x) = x^2$

(b) $U(x) = x$

(c) $U(x) = \sqrt{x}$

(d) None of the above

Part B. Assume Agent 1 has utility U_1 and Agent 2 has utility U_2 . If

$$U_1 = k_1 U_2 + k_2, \quad k_1 > 0, \quad k_2 \in \mathbf{R},$$

do Agents 1 and 2 have the same preferences?

Problem 2: Adventures of PacBaby

PacBaby just found a \$100 note — it is the only thing she owns. Monsters are kind enough not to kill PacBaby, but whenever they find her, they steal all her money. The probability that the monsters find PacBaby is 20% (i.e., 0.2).

PacBaby's utility function is

$$U(x) = \log(1 + x),$$

where $\log$ denotes the natural logarithm and x is her monetary wealth. Thus,

$$U(100) = \log(101) \quad \text{and} \quad U(0) = \log(1 + 0) = 0.$$

- (a) What is PacBaby's expected utility without insurance?
- (b) PacIncome offers theft insurance: for a premium of \$30, PacBaby will be reimbursed \$70 if monsters steal her money. After paying the premium, if no theft occurs, PacBaby holds \$70; if theft occurs, she also ends up with \$70 (the reimbursement). What is PacBaby's expected utility if she buys insurance? Should she buy it to maximize expected utility?
- (c) From PacIncome's point of view, what is the expected *monetary* value of selling this insurance?

Problem 3: Allais Paradox

The Allais paradox (Allais, 1953) is a well-known problem potentially suggesting that humans are “predictably irrational” (Ariely, 2009)¹. People are given a choice between lotteries A and B and then between C and D, which have the following prizes:

A: 80% chance of \$4000 C: 20% chance of \$4000
 B: 100% chance of \$3000 D: 25% chance of \$3000

Most people consistently prefer B over A (i.e., taking the sure payoff), and C over D (taking the higher EMV).

- a) Show that the normative analysis (i.e., describing how a rational agent should act) disagrees. [Hint: Set $U(\$0) = 0$; show that the preferences between A, B and C, D are opposites, hence a contradiction.]
- b) Prove that the judgments $B \succ A$ and $C \succ D$ in the above Allais paradox violate the axiom of substitutability. [Hint: You may wish to consider using the axiom of decomposability.]

¹For a possible explanation of such a paradox, refer to page 620 of AIMA textbook.